

Day 03: Operators and Conditional Statements

Learning Objectives

After completing this lesson, you will be able to:

Understand Python operators

Perform arithmetic calculations

Compare values

Use logical operators

Write decision-making programs using if, if-else, and elif

1. What are Operators?

Operators are special symbols used to perform operations on variables and values.

Example:

```
a = 10
```

```
b = 5
```

```
print(a + b)
```

Output:

```
15
```

2. Types of Operators

Arithmetic Operators

Operator

Description

Example

+

Addition

```
10 + 5
```

-

Subtraction

10 - 5

*

Multiplication

10 * 5

/

Division

10 / 5

%

Modulus

10 % 3

//

Floor Division

10 // 3

**

Exponent

2 ** 3

Example:

a = 20

b = 6

print(a+b)

print(a-b)

print(a*b)

print(a/b)

print(a%b)

3. Comparison Operators

Comparison operators compare two values.

Operator

Meaning

==

Equal to

!=

Not Equal

>

Greater Than

<

Less Than

>=

Greater Than or Equal

<=

Less Than or Equal

Example:

x = 20

y = 15

```
print(x > y)
```

```
print(x == y)
```

Output:

True

False

4. Logical Operators

Logical operators combine conditions.

and

Returns True only if both conditions are True.

Example:

```
age = 20
```

```
print(age > 18 and age < 25)
```

or

Returns True if at least one condition is True.

Example:

```
print(True or False)
```

not

Reverses the result.

Example:

```
print(not True)
```

Output:

False

5. Assignment Operators

Example:

```
x = 10
```

```
x += 5
```

```
print(x)
```

```
x -= 2
```

```
print(x)
```

6. What are Conditional Statements?

Conditional statements allow Python to make decisions.

Python provides:

if

if-else

elif

Nested if

7. if Statement

Syntax:

if condition:

 statement

Example:

age = 20

if age >= 18:

 print("Eligible to Vote")

8. if-else Statement

Example:

marks = 40

if marks >= 35:

 print("Pass")

else:

 print("Fail")

9. elif Statement

Example:

marks = 85

if marks >= 90:

 print("Grade A+")

elif marks >= 75:

```
    print("Grade A")

elif marks >= 60:

    print("Grade B")

else:

    print("Grade C")
```

10. Nested if

Example:

```
age = 22
```

```
if age >= 18:

    if age <= 60:

        print("Eligible")
```

11. Practice Programs

Program 1

Check Even or Odd

```
num = int(input("Enter Number: "))
```

```
if num % 2 == 0:

    print("Even")

else:

    print("Odd")
```

Program 2

Find Largest Number

```
a = 10
```

```
b = 20
```

```
if a > b:
```

```
print(a)
```

```
else:
```

```
print(b)
```

Program 3

Positive or Negative

```
num = int(input("Enter Number: "))
```

```
if num >= 0:
```

```
    print("Positive")
```

```
else:
```

```
    print("Negative")
```